

A Fixed-Section Counterexample for the Infinite-Dimensional QR Algorithm

Agent lane 5, run `fa_banach_001`

2026-07-19

Source and Target

The source paper is M. J. Colbrook and A. C. Hansen, *On the infinite-dimensional QR algorithm*, arXiv:2011.08172 [1]. In the concluding remarks, the authors write that their examples lead them to conjecture that, for normal operators, the IQR algorithm recovers the extreme parts of the essential spectrum when the set of extremal points of the essential spectrum has size one. In the source TeX this is at lines 2167–2171; in the locally compiled source PDF it is on page 42.

eigenvalues of distinct magnitude. These results present the first such algorithms that tackle these problems with error control.

Finally, we demonstrated that the IQR algorithm can be implemented both in theory and in practice. We demonstrated the convergence theorems in Section 3 as well as some examples (normal and non-normal) where the extremal parts of the essential spectrum also appear to be recovered. Based on this, we conjecture that there may be a large class of operators for which the IQR algorithm converges to the extreme parts of the essential spectrum.⁴ In particular, we conjecture that this holds for normal operators if the set of extremal points of the essential spectrum has size one. However, an example was given where convergence to the essential spectrum was only algebraic $O(n^{-\alpha})$ as $n \rightarrow \infty$ as opposed to the linear convergence rate $O(r^n)$ to the discrete spectrum/eigenvalues. It was also demonstrated that the truncations of the IQR algorithm have spectra agreeing with periodic boundary conditions for a range of operators in the class of “pseudoergodic” NSA random operators. In some cases, the algorithm performed much better than standard finite section methods. We should stress that, as in the case of the finite section method, for fixed n and $m \rightarrow \infty$, the output $\sigma(P_m Q_n^* T Q_n|_{P_m})$ will in general still suffer from the spectral pollution phenomenon and in some cases not recover the full spectrum. This was apparent in the numerical examples and is likely to hold true for many operators even when taking a mixture of double limits $m, n \rightarrow \infty$. However, examples of Laurent operators were given where it appears $\sigma(P_m Q_n^* T Q_n|_{P_m})$ converges to the spectrum as $m \rightarrow \infty$ for fixed $n > 0$ but not $n = 0$. We hope that the algorithm’s potential use in sifting out spectral pollution/complying

The counterexample below addresses the natural fixed-finite-section reading of that sentence: after choosing a sufficiently large finite m , the limit $n \rightarrow \infty$ of the spectra of

$$P_m Q_n^* T Q_n|_{P_m \mathcal{H}}$$

should recover the unique extremal point of the essential spectrum. The example shows that no finite m , however large, can do this in general.

Proof Intuition

The IQR iterates do not create spectral support in the first m columns. The source paper records the identity

$$\text{span}\{Q_n e_j : 1 \leq j \leq m\} = \text{span}\{T^m e_j : 1 \leq j \leq m\}.$$

If the first m basis vectors sit inside a finite reducing subspace F of T , then the IQR m -section is only a compression of T to a subspace of F . Thus it can never see spectral points that occur only as an asymptotic tail of later reducing blocks. Taking non-diagonal 2×2 blocks whose eigenvalues increase to 1 gives $\sigma_{\text{ess}}(T) = \{1\}$, but each fixed finite leading section remains separated from 1.

Counterexample

Theorem 1. *There exists an invertible normal operator $T \in \mathcal{B}(\ell^2(\mathbb{N}))$ such that:*

1. T is self-adjoint, positive, and non-diagonal in the canonical basis;
2. $\sigma_{\text{ess}}(T) = \{1\}$, so the essential spectrum has exactly one point of maximal modulus;
3. T has no eigenvalue of modulus greater than $r_{\text{ess}}(T) = 1$;
4. for every $m \in \mathbb{N}$ there is a number $\varepsilon_m > 0$ such that, for every IQR Q -sequence (Q_n) of T and every $n \in \mathbb{N}$,

$$\text{dist}(1, \sigma(P_m Q_n^* T Q_n|_{P_m \ell^2})) \geq \varepsilon_m.$$

Consequently the unique extremal point $1 \in \sigma_{\text{ess}}(T)$ is not recovered by the fixed- m , $n \rightarrow \infty$ IQR sections for any sufficiently large finite m .

Proof. For $k \geq 1$, set

$$a_k = 1 - \frac{1}{2k}, \quad b_k = 1 - \frac{1}{2k+1}.$$

Then

$$0 < a_1 < b_1 < a_2 < b_2 < \cdots < 1, \quad a_k, b_k \rightarrow 1.$$

Let

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and define

$$B_k = U^* \text{diag}(a_k, b_k) U = \begin{pmatrix} (a_k + b_k)/2 & (a_k - b_k)/2 \\ (a_k - b_k)/2 & (a_k + b_k)/2 \end{pmatrix}.$$

Each B_k is a non-diagonal real symmetric positive matrix with eigenvalues a_k and b_k . With respect to the canonical decomposition

$$\ell^2(\mathbb{N}) = \bigoplus_{k=1}^{\infty} \text{span}\{e_{2k-1}, e_{2k}\},$$

define

$$T = \bigoplus_{k=1}^{\infty} B_k.$$

Then T is bounded, self-adjoint, positive, and invertible, since $\inf_k a_k = a_1 = 1/2$. It is not diagonal in the canonical basis because $(a_k - b_k)/2 \neq 0$ for every k .

Moreover,

$$T - I = \bigoplus_{k=1}^{\infty} (B_k - I_2)$$

is compact, because $\|B_k - I_2\| = 1 - a_k \rightarrow 0$. In the Calkin algebra, the image of T is therefore the image of I , and hence

$$\sigma_{\text{ess}}(T) = \{1\}.$$

The point 1 is the unique point of the essential spectrum and the unique point of maximal modulus in it. The eigenvalues of T are exactly the numbers a_k, b_k , all of which are strictly smaller than 1. Thus there are no eigenvalues with modulus greater than $r_{\text{ess}}(T) = 1$.

Fix $m \in \mathbb{N}$ and let $K = \lceil m/2 \rceil$. Put

$$F_K = \bigoplus_{k=1}^K \text{span}\{e_{2k-1}, e_{2k}\}.$$

This is a finite-dimensional reducing subspace for T , and $\text{span}\{e_1, \dots, e_m\} \subset F_K$. Therefore

$$T^n \text{span}\{e_1, \dots, e_m\} \subset F_K \quad (n \geq 0).$$

For the IQR Q -sequence, the source paper's identity

$$\text{span}\{Q_n e_j : 1 \leq j \leq m\} = \text{span}\{T^n e_j : 1 \leq j \leq m\}$$

follows from $T^n = Q_n R_n$ and the fact that R_n is upper triangular with non-zero diagonal. Hence

$$V_{m,n} := \text{span}\{Q_n e_1, \dots, Q_n e_m\} \subset F_K \quad (n \geq 0).$$

The matrix $P_m Q_n^* T Q_n|_{P_m \ell^2}$ is the matrix of the compression of the self-adjoint operator T to the m -dimensional subspace $V_{m,n}$, written in the orthonormal basis $Q_n e_1, \dots, Q_n e_m$. Since $V_{m,n} \subset F_K$, the min-max principle gives

$$\sigma(P_m Q_n^* T Q_n|_{P_m \ell^2}) \subset [a_1, b_K].$$

Consequently

$$\text{dist}(1, \sigma(P_m Q_n^* T Q_n|_{P_m \ell^2})) \geq 1 - b_K = \frac{1}{2K+1} > 0$$

for every n . Taking $\varepsilon_m = 1/(2\lceil m/2 \rceil + 1)$ proves the claim. $\square$

Interpretive Scope

This packet does not rule out weaker double-limit statements in which m is allowed to tend to infinity after, before, or together with n . Indeed, in the example above, $b_{\lceil m/2 \rceil} \rightarrow 1$ as $m \rightarrow \infty$. What fails is the fixed finite-section conclusion suggested by the source phrase “in the limit $n \rightarrow \infty$ for large enough m ”: no finite m has n -iterates whose m -section can reach the unique essential extremal point.

Verification Notes

The proof is analytic and uses no numerical hypothesis. The accompanying script `code/verify_block_counterexample.py` checks finite block eigenvalues, the non-diagonal form of the blocks, and the predicted positive gaps

$$1 - b_{\lceil m/2 \rceil}.$$

It is only a sanity check; the proof above is independent of floating-point computation.

Novelty Check

Before promotion, the run indexes were searched for arXiv:2011.08172, *On the infinite-dimensional QR algorithm*, IQR, QR algorithm, QR iteration, and Hessenberg; no duplicate solution packet or attempt for this paper was found. A bounded web search on 2026-07-19 for exact title/id phrases and the phrases “IQR algorithm”, “normal operators”, “essential spectrum”, and “extremal points” found the source paper but no later answer or counterexample to this exact conjectural fixed-section statement. The novelty confidence is therefore moderate: the construction is elementary, but no pre-existing answer was found in the checked sources.

References

References

- [1] M. J. Colbrook and A. C. Hansen, *On the infinite-dimensional QR algorithm*, Numerische Mathematik 143 (2019), 17–83; arXiv:2011.08172.